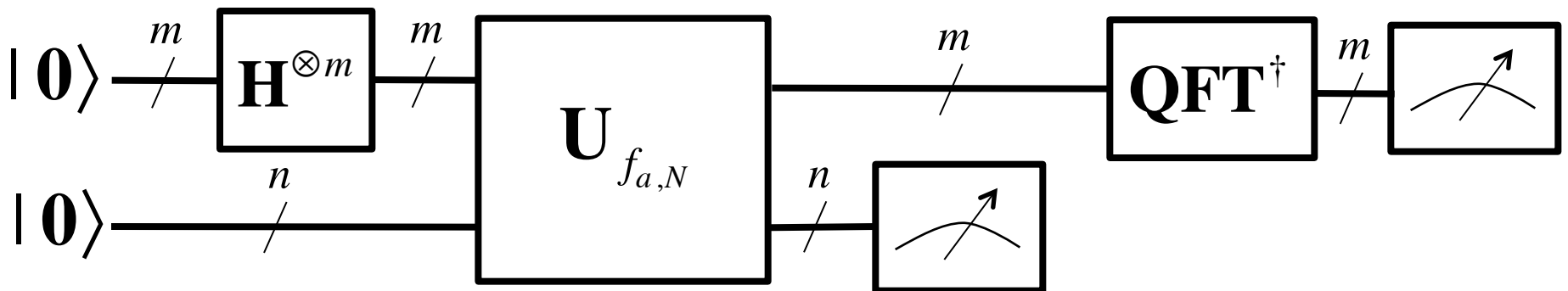
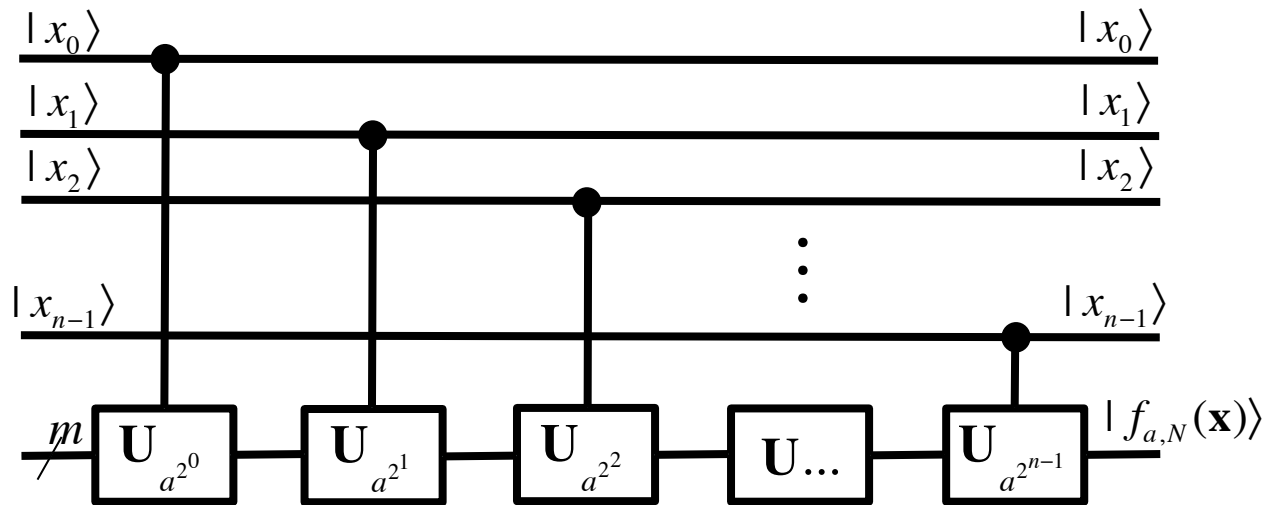


Shor's Factoring Algorithm



Shor's Factoring Algorithm

- Factoring Composite Numbers Very Important and Used for Security (Encryption)
- Method Reduces Factoring Problem to Finding Period of Function
- Deterministic Polynomial Algorithm Exists for Determining if a Value is Prime
 - Agrawal, Kayal, and Saxena, 2004
- Assumed Number is Already Checked for Primality

Modular Operation

- Modular Arithmetic
- Notation Where k, j , and r are Integers

$$|k|_j = r$$

- Denotes the Congruence:

$$k \equiv r \pmod{j} \Rightarrow k \pmod{j} = r \pmod{j}$$

- Such that:

$$j > 0 \qquad 0 \leq r \leq j - 1$$

- Examples:

$$|7|_{15} = 7 \qquad |199|_{15} = 4 \qquad |23374|_{371} = 1$$

$$|99|_{15} = 9 \qquad |5317|_{371} = 123 \qquad |1446|_{371} = 333$$

Euclid's Algorithm

- Method for Computing the Greatest Common Divisor (GCD)
- GCD of Two Numbers is the Largest Number that Divides Both WITH a Zero-Valued Remainder
- Principle is GCD of Two Numbers Does Not Change if Smaller Number is Subtracted from Larger Number:

$$GCD(252,105) = 21$$

$$\begin{aligned} GCD(252,105) &= GCD(252 - 105,105) = GCD(147,105) = 21 \\ &= GCD(147,105) = GCD(147 - 105,105) = GCD(42,105) \end{aligned}$$

$$GCD(42,105) = GCD(63,42) = GCD(42,21) = GCD(21,0) = 21$$

CoPrimes

- Coprime Definition:
- Two Numbers a and b are Coprime if:
$$GCD(a,b) = 1$$
- When Searching for Factor of Number N :
 - Randomly Choose some value a Where $a < N$
 - Invoke Euclid's Algorithm for $GCD(a, N)$
 - If $GCD(a, N) \neq 1$, Then Factor of N is Found
 - If $GCD(a, N) = 1$, Then a is Coprime to N and can be Used
- Next, we Find Powers of a Modulo N :
$$| a^0 |_N, | a^1 |_N, | a^2 |_N, | a^3 |_N, \dots$$

Modular Powers Function

- Finding Powers of a Modulo N Equivalent to Finding Values of Function:

$$f_{a,N}(x) = a^x \pmod{N} = |a^x|_N$$

- **EXAMPLE:** $N=15$ and $a=2$:

x	0	1	2	3	4	5	6	7	8	9	10	11	12	...
$f_{2,15}(x)$	1	2	4	8	1	2	4	8	1	2	4	8	1	...

- **EXAMPLE:** $N=15$ and $a=4$:

[illegible]

Modular Powers Function

- Finding Powers of a Modulo N Equivalent to Finding Values of Function:

$$f_{a,N}(x) = a^x \pmod{N} = |a^x|_N$$

- EXAMPLE: $N=15$ and $a=13$:

x	0	1	2	3	4	5	6	7	8	9	10	11	12	...
$f_{13,15}(x)$	1	13	4	7	1	13	4	7	1	13	4	7	1	...

Modular Powers Function

- Useful Identities:

if $a \equiv a' \pmod{N}$ and $b \equiv b' \pmod{N}$

then $a \times b \equiv a' \times b' \pmod{N}$

$$|a \times b|_N = ||a|_N \times |b|_N|_N$$

$$|a^x|_N = |a^{x-1} \times a|_N = ||a^{x-1}|_N \times |a|_N|_N$$

- Since $a < N$ and $|a|_N = a$, Above Reduces to:

$$|a^x|_N = ||a^{x-1}|_N \times a|_N$$

- This Identity Allows Larger Values to be Used

Modular Powers Function

$$f_{a,N}(x) = a^x \pmod{N} = |a^x|_N$$

- EXAMPLE: $N=371$ and $a=2$:

x	0	1	2	3	4	5	6	7	...	78	...	155	156	157	158	...
$f_{2,371}(x)$	1	2	4	8	16	32	64	128	...	211	...	186	1	2	4	...

- EXAMPLE: $N=371$ and $a=6$:

x	0	1	2	3	4	5	6	7	...	13	...	25	26	27	28	...
$f_{6,371}(x)$	1	6	36	216	183	356	281	202	...	370	...	62	1	6	36	...

- EXAMPLE: $N=371$ and $a=24$:

x	0	1	2	3	4	5	6	7	...	39	...	77	78	79	80	...
$f_{24,371}(x)$	1	24	205	97	102	222	134	248	...	160	...	201	1	24	205	...

Modular Powers Function

- These Functions are Periodic
- We Only Need Period of Function
- Period: Find Smallest $x > 0$ Such That:

$$f_{a,N}(x) = a^x \pmod{N} = |a^x|_N = 1$$

- Number Theory Theorem: For any coprime $a \leq N$, the function $f_{a,N}$ will evaluate to 1 for some $x < N$. After, $f_{a,N}$ evaluates to 1, the sequence of function values repeats.
- If $f_{a,N}(x) = 1$, then

$$f_{a,N}(x+1) = f_{a,N}(1) \qquad f_{a,N}(x+s) = f_{a,N}(s)$$

Finding the Period

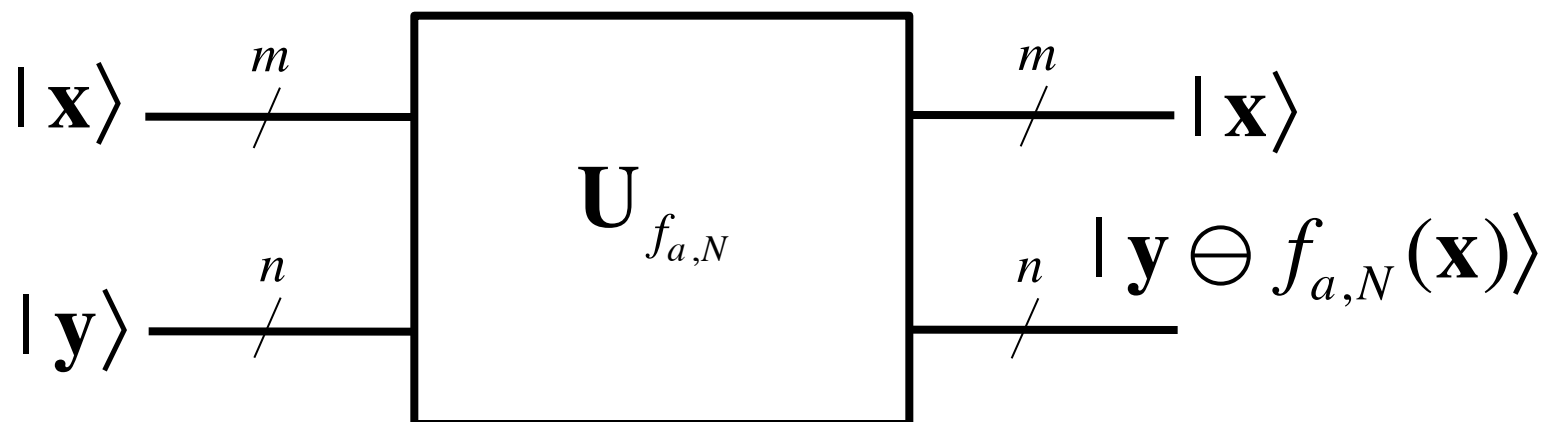
- For Small Numbers (15, 371, 247) Easy to Compute the Period
- Large Numbers with Hundreds of Digits are Beyond Capability of Classical Computers
- Use Quantum Computer with Qubit Superposition to Calculate $f_{a,N}(x)$ for all Needed x
- Must First Synthesize $f_{a,N}(x)$ into a Quantum Cascade

Period Finding Quantum Circuit

- Number of Qubits Required
 - $f_{a,N}$ Always Evaluates to Value Less Than N
 - Need $n=\log_2(N)$ Qubits to Represent Function Value
 - Need to Evaluate $f_{a,N}$ for at Least First N^2 Values of x , Thus Need $m=\log_2(N^2)=2\log_2(N)=2n$ Qubits for x Values
- Quantum Circuit Represented by Operator:

$$\mathbf{U}_{f_{a,N}}$$

Period Finding Quantum Circuit



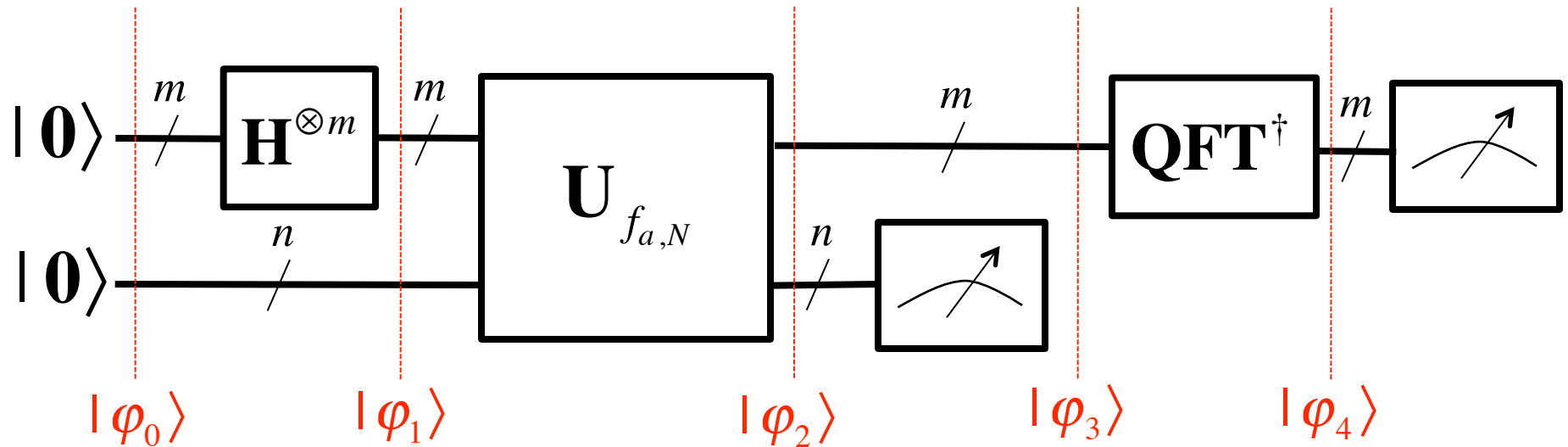
$$|\mathbf{x}, \mathbf{y}\rangle \mapsto |\mathbf{x}, \mathbf{y} \ominus f_{a,N}(\mathbf{x})\rangle = |\mathbf{x}, |\mathbf{y} \oplus a^{\mathbf{x}}|_N\rangle$$

$$n = \lceil \log_2(N) \rceil$$

$$m = \lceil \log_2(N^2) \rceil = \lceil 2\log_2(N) \rceil = 2n$$

- Discussion of Circuit Structure Postponed for Now

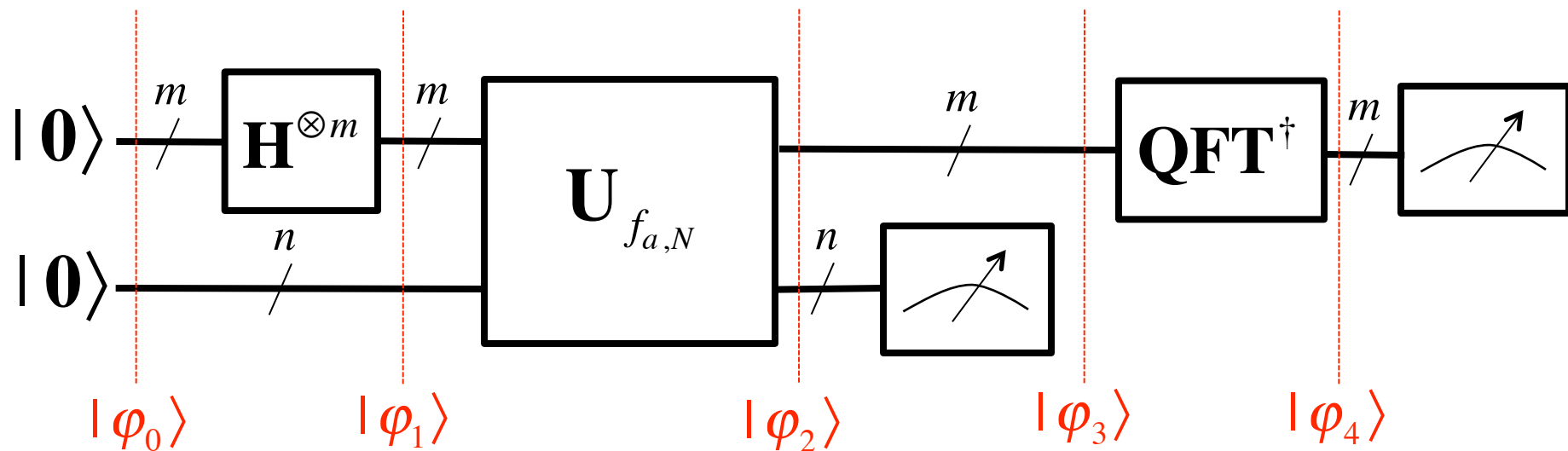
Quantum Circuit



- Evaluate All Input Simultaneously Through Superposition
- Quantum Circuit Transfer Matrix:

$$(Measure_m \otimes I_n)(QFT_m^\dagger \otimes I_n)(I_m \otimes Measure_n)U_{f_{a,N}}(H^{\otimes m} \otimes I_n)|0_m, 0_n\rangle$$

Quantum Circuit



$$|\varphi_0\rangle = |\mathbf{0}_m, \mathbf{0}_n\rangle = |\mathbf{0}_m\rangle \otimes |\mathbf{0}_n\rangle$$

$$|\varphi_1\rangle = (\mathbf{H}^{\otimes m} \otimes \mathbf{I}_n) |\mathbf{0}_m, \mathbf{0}_n\rangle = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, \mathbf{0}_n\rangle}{\sqrt{2^m}}$$

$$|\varphi_2\rangle = (\mathbf{H}^{\otimes m} \otimes \mathbf{I}_n) \mathbf{U}_{f_{a,N}} |\mathbf{0}_m, \mathbf{0}_n\rangle = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, f_{a,N}(\mathbf{x})\rangle}{\sqrt{2^m}} = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, |a^{\mathbf{x}}|_N\rangle}{\sqrt{2^m}}$$

Example Calculation 1

$$|\varphi_2\rangle = (\mathbf{H}^{\otimes m} \otimes \mathbf{I}_n) U_{f_{a,N}} |\mathbf{0}_m, \mathbf{0}_n\rangle = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, f_{a,N}(\mathbf{x})\rangle}{\sqrt{2^m}} = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, |a^{\mathbf{x}}|_N\rangle}{\sqrt{2^m}}$$

- Assume $N=15$ and $a=13$:

$$n = \lceil \log_2(N) \rceil = \lceil \log_2(15) \rceil = 4$$

$$m = \lceil \log_2(15^2) \rceil = 2 \lceil \log_2(15) \rceil = 2 \times 4 = 8$$

$$|\varphi_2\rangle = \frac{|0,1\rangle + |1,13\rangle + |2,4\rangle + |3,7\rangle + |4,1\rangle + \dots + |254,4\rangle + |255,7\rangle}{\sqrt{256}}$$

x	0	1	2	3	4	5	6	7	8	9	10	11	12	...
$f_{13,15}(x)$	1	13	4	7	1	13	4	7	1	13	4	7	1	...

Example Calculation 2

$$|\varphi_2\rangle = (\mathbf{H}^{\otimes m} \otimes \mathbf{I}_n) U_{f_{a,N}} |\mathbf{0}_m, \mathbf{0}_n\rangle = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, f_{a,N}(\mathbf{x})\rangle}{\sqrt{2^m}} = \frac{\sum_{\mathbf{x} \in \{0,1\}^m} |\mathbf{x}, |a^{\mathbf{x}}|_N\rangle}{\sqrt{2^m}}$$

- Assume $N=371$ and $a=24$:

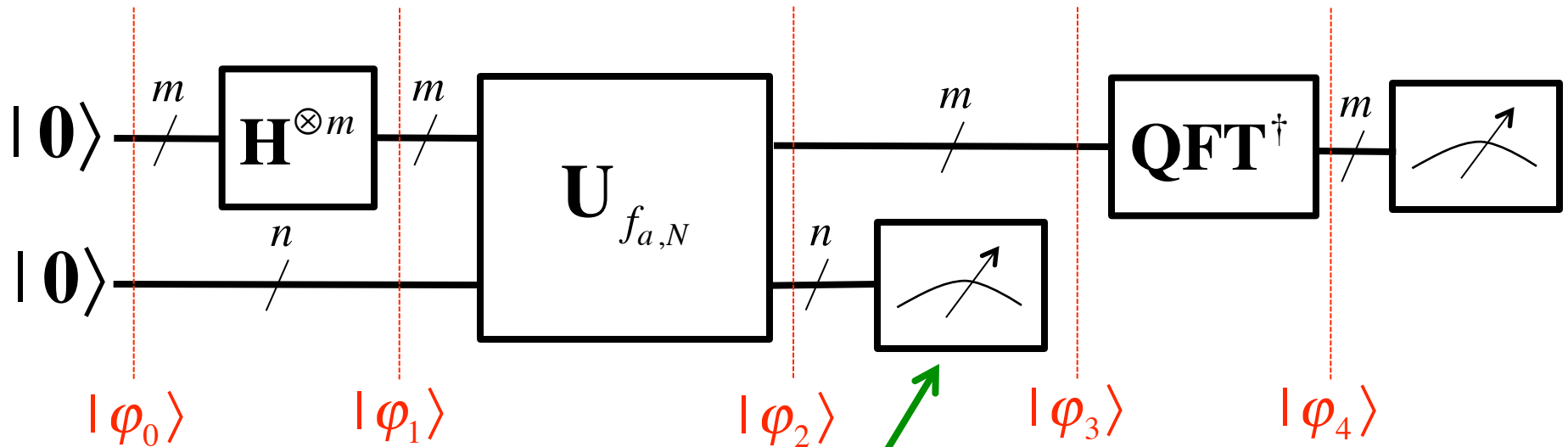
$$n = \lceil \log_2(N) \rceil = \lceil \log_2(371) \rceil = 9$$

$$m = \lceil \log_2(371^2) \rceil = 2 \lceil \log_2(371) \rceil = 2 \times 9 = 18$$

$$|\varphi_2\rangle = \frac{|0,1\rangle + |1,24\rangle + |2,205\rangle + |3,97\rangle + |4,102\rangle + \dots + |2^{18}-1, |24^{2^{18}-1}|_{371}\rangle}{\sqrt{2^{18}}}$$

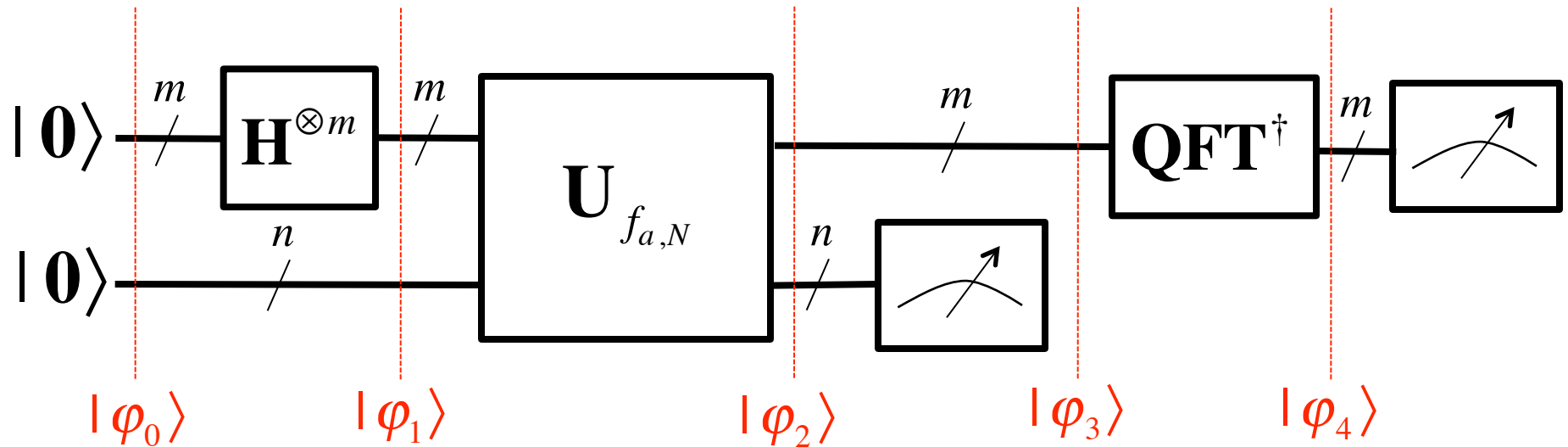
x	0	1	2	3	4	5	6	7	...	39	...	77	78	79	80	...
$f_{24,371}(x)$	1	24	205	97	102	222	134	248	...	160	...	201	1	24	205	...

Quantum Circuit



- Measure Bottom n Qubits of
- Bottom n Qubits are in State of Superposition Before Measurement
- Assume We Measure: $|a^{\bar{x}}|_N$
- For Some Particular Bitstring: $\bar{x}$

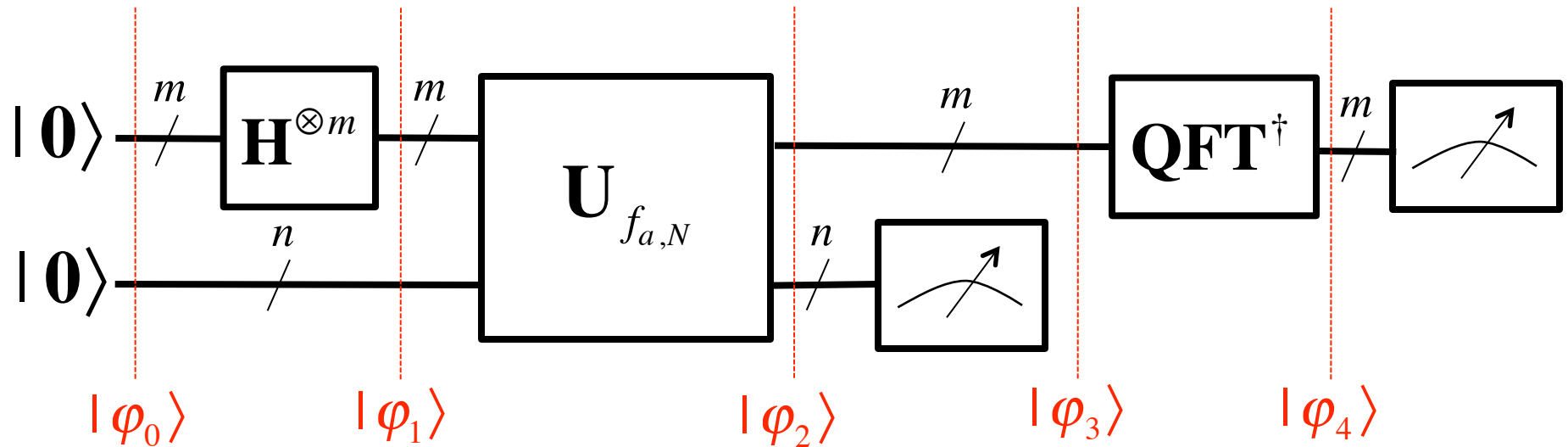
Quantum Circuit



- Assume We Measure: $|a^{\bar{x}}|_N$
- For Some Particular Bitstring: $\bar{x}$
- Since $f_{a,N}$ is Periodic: $a^{\bar{x}} \equiv |a^{\bar{x}+r}|_N$ And $a^{\bar{x}} \equiv |a^{\bar{x}+2r}|_N$
- For any $s \in \mathbb{Z}$

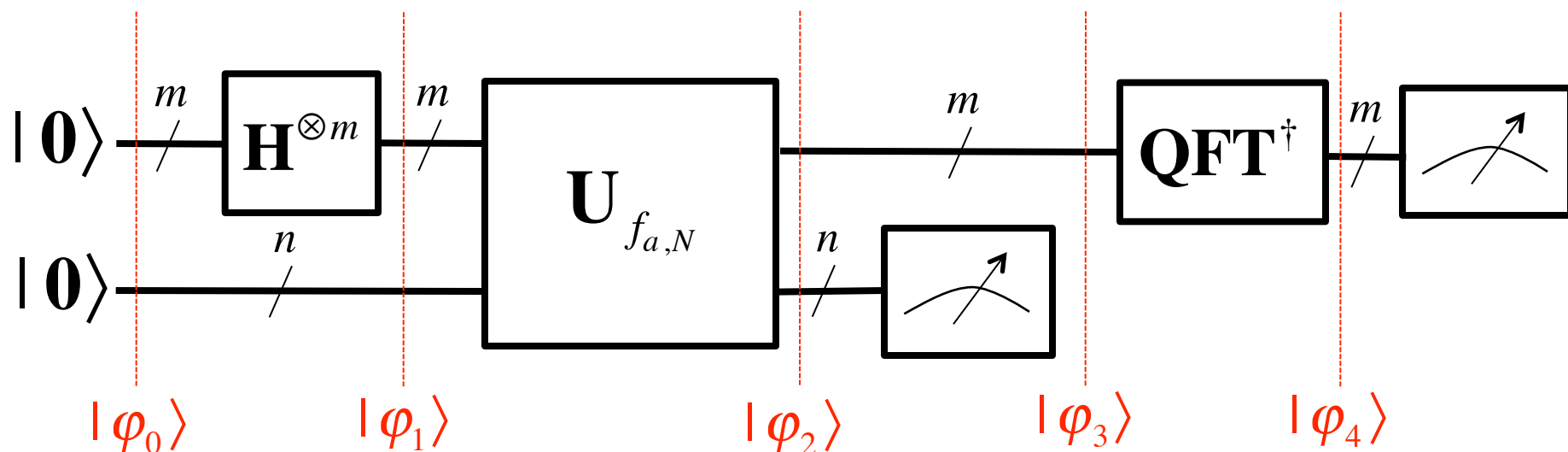
$$a^{\bar{x}} \equiv |a^{\bar{x}+sr}|_N$$

Quantum Circuit



- There are 2^m Superpositions in $|\varphi_2\rangle$
- The Number of Superpositions that have $|a^{\bar{x}}|_N$ as the Result are $\left\lfloor \frac{2^m}{r} \right\rfloor$
- This Result is used in the Expression for $|\varphi_3\rangle$

Quantum Circuit

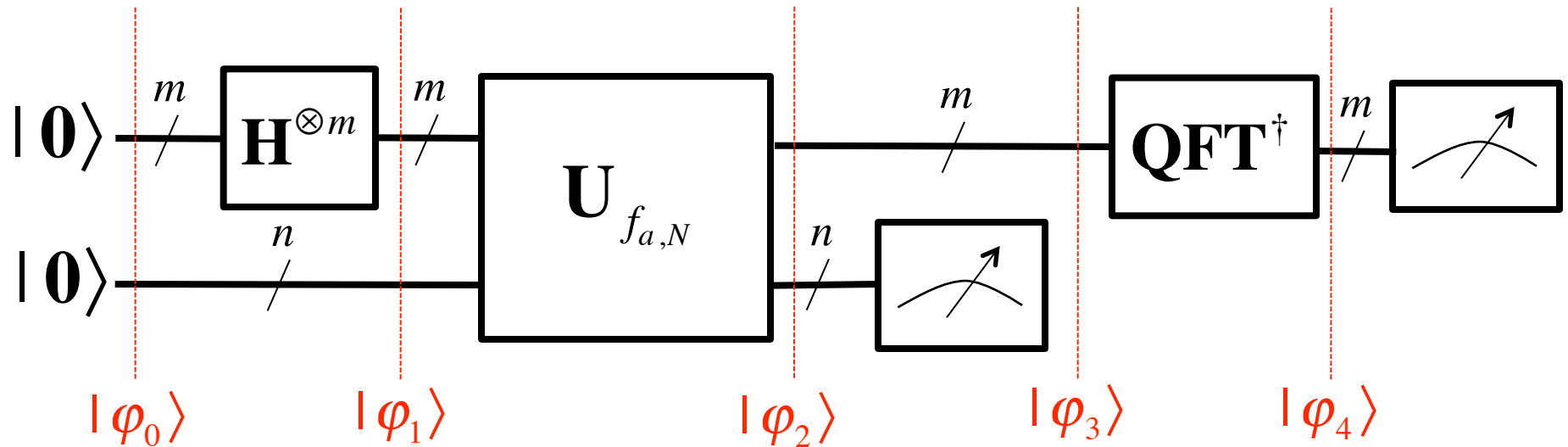


$$|\varphi_3\rangle = \frac{\sum_{a^{\bar{x}} \equiv |a^{\bar{x}}|_N} |x, |a^{\bar{x}}|_N\rangle}{\left\lfloor \frac{2^m}{r} \right\rfloor} = \frac{\sum_{j=0}^{2^m/(r-1)} |t_0 + jr, |a^{\bar{x}}|_N\rangle}{\left\lfloor \frac{2^m}{r} \right\rfloor}$$

- t_0 is the Offset, the First Occurrence of:

$$a^{t_0} \equiv |a^{\bar{x}}|_N$$

Quantum Circuit



- t_0 is Called the Offset
- $|\varphi_3\rangle$ Stage Employs Entanglement
- Top m and Bottom n Qubits are Entangled Such That When Top n are Measured, the Bottom m Retain Their State

Example Calculation 1

- Recall Earlier Result:

$$|\varphi_2\rangle = \frac{|0,1\rangle + |1,13\rangle + |2,4\rangle + |3,7\rangle + |4,1\rangle + \dots + |254,4\rangle + |255,7\rangle}{\sqrt{256}}$$

- Assume that After Measurement of Bottom $n=4$ Qubits, the Value 7 is Obtained:

$$\bar{\mathbf{x}} = 0111 = 7$$

- The Quantum State Becomes:

$$|\varphi_3\rangle = \frac{|3,7\rangle + |7,7\rangle + |11,7\rangle + |15,7\rangle + \dots + |251,7\rangle + |255,7\rangle}{\left[\frac{256}{4} \right]}$$

Example Calculation 2

- Recall Earlier Result:

$$|\varphi_2\rangle = \frac{|0,1\rangle + |1,24\rangle + |2,205\rangle + |3,97\rangle + |4,102\rangle + \dots + |2^{18}-1, |24^{2^{18-1}}|_{371}\rangle}{\sqrt{2^{18}}}$$

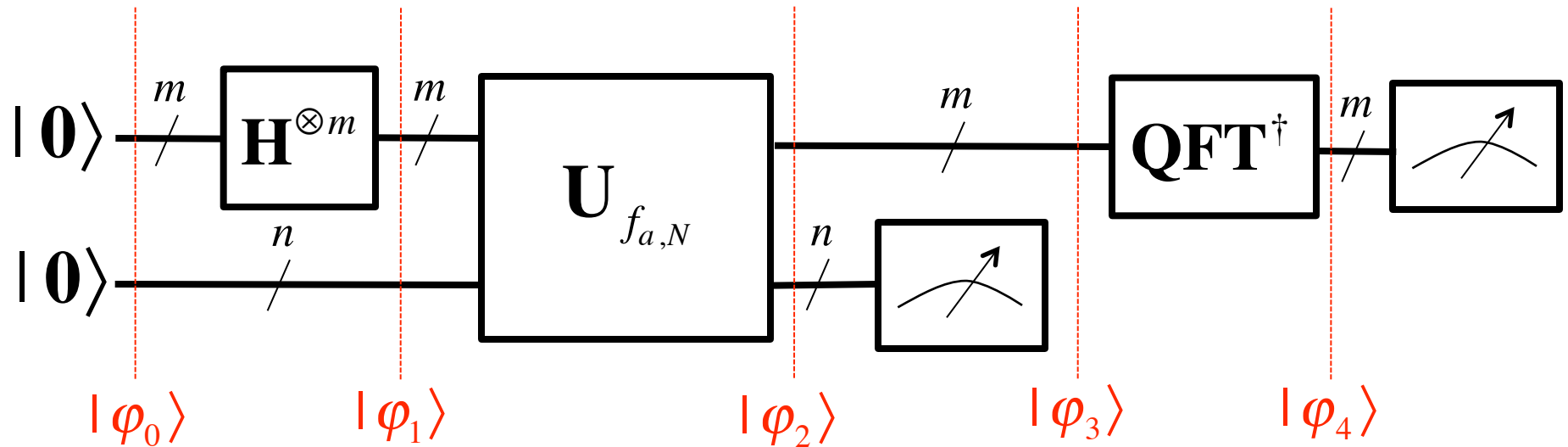
- Assume that After Measurement of Bottom $n=9$ Qubits, the Value 222 is Obtained:

$$\bar{\mathbf{x}} = 011011110 = 222 = |24^5|_{371}$$

- The Quantum State Becomes:

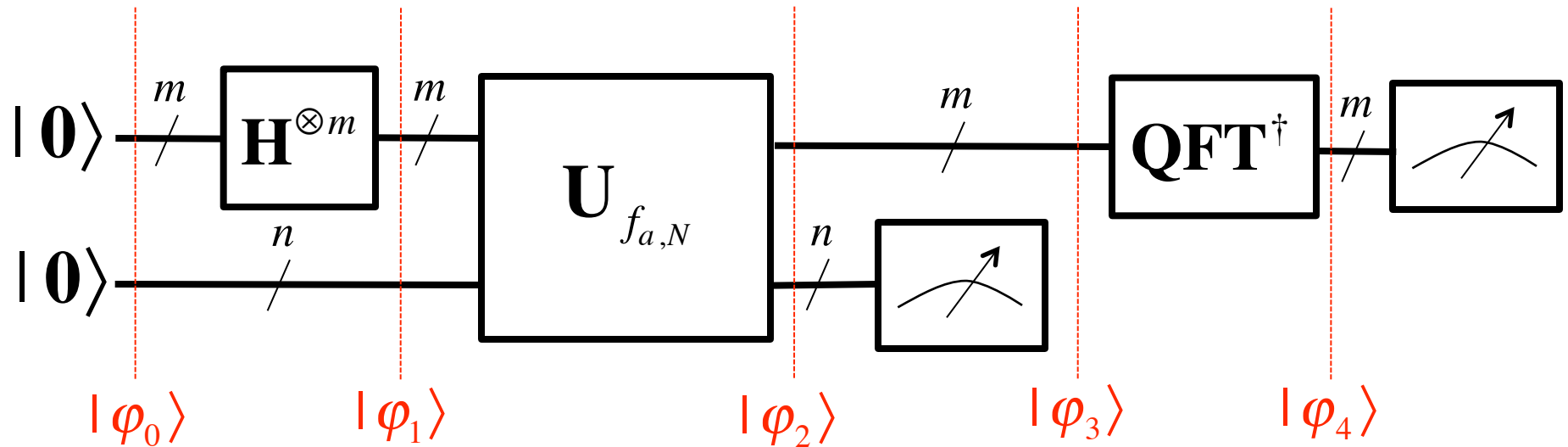
$$|\varphi_3\rangle = \frac{|5,222\rangle + |83,222\rangle + |161,222\rangle + |239,222\rangle + \dots}{\left[\frac{2^{18}}{78} \right]}$$

Quantum Circuit



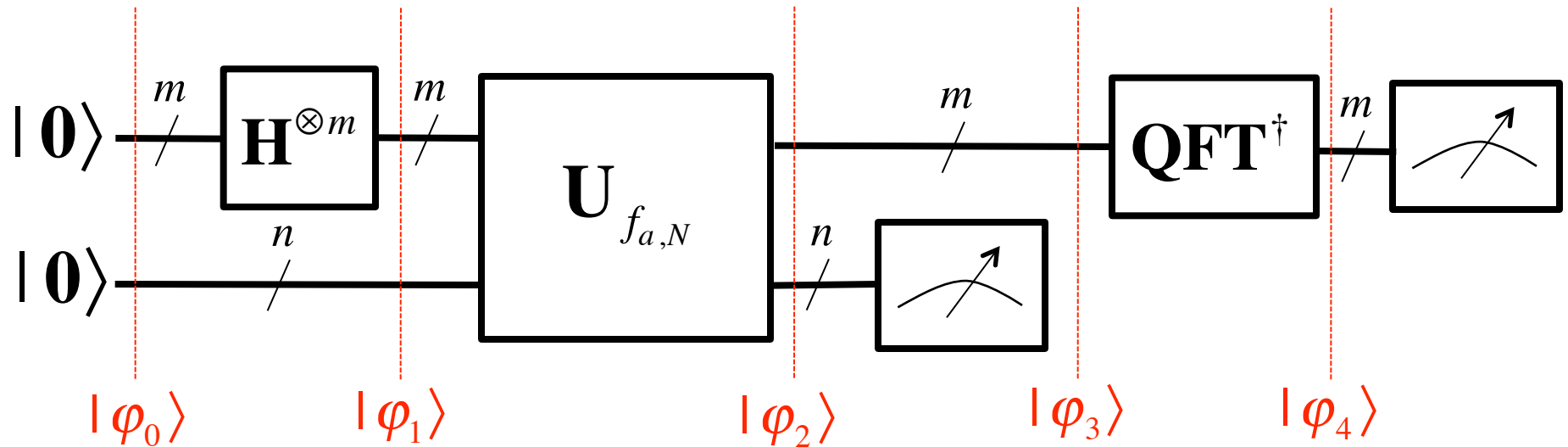
- $|\varphi_4\rangle$ Step of the Quantum Part of Algorithm is Application of the Inverse Quantum Fourier Transform
- Final Step of Algorithm Measures the Top m Qubits

Quantum Circuit



- Make Simplifying Assumption that r Evenly Divides into 2^m
- Shor's Actual Algorithm Does Not Make this Assumption

Quantum Circuit



- With Simplifying Assumption, We Measure:

$$x = \frac{\lambda 2^m}{r}$$

- Where λ is Some Whole Number

Quantum Circuit

- Known Values After Measurement are 2^m and x
- Dividing Whole Number x by 2^m Yields

$$\frac{x}{2^m} = \frac{\lambda 2^m}{r 2^m} = \frac{\lambda}{r}$$

- $\frac{\lambda}{r}$ is Reduced to an Irreducible Fraction and Denominator Then Becomes the Sought After r Value (the period)
- Without Simplifying Assumption, Process is Repeated and Results are Analyzed to Obtain r

Using Period to Get Factors

- We now Know the Period of $f_{a,N}$ for Some Value of a
- Number Theory Theorem States that for the Majority of a Values, r is an Even Number
- If it Turns Out that r is Odd, We Throw the Result Out and Try Again by Choosing Another a Value
- Once Even r is Found, We Have:

$$a^r \equiv |1|_N$$

Using Period to Get Factors

- Subtracting 1 From Both Sides of the Congruence Yields:

$$a^r - 1 \equiv |0|_N$$

$$N \mid (a^r - 1)$$

- Using the Facts:

$$1 = 1^2 \qquad x^2 - y^2 = (x + y)(x - y)$$

- Results in:

$$N \mid (a^r - 1) = N \mid (\sqrt{a^r} + 1)(\sqrt{a^r} - 1) = N \mid \left(a^{\frac{r}{2}} + 1\right) \left(a^{\frac{r}{2}} - 1\right)$$

Using Period to Get Factors

- Since r is Even, Exponent Yields a Whole Number

$$N \mid (a^r - 1) = N \mid (\sqrt{a^r} + 1)(\sqrt{a^r} - 1) = N \mid \left(a^{\frac{r}{2}} + 1\right)\left(a^{\frac{r}{2}} - 1\right)$$

- Any Factor of N is Also a Factor of

$$\left(a^{\frac{r}{2}} + 1\right) \text{ or } \left(a^{\frac{r}{2}} - 1\right)$$

- Can Employ Classical Euclid's Algorithm to Search for Factor of N

$$\text{GCD}\left(\left(a^{\frac{r}{2}} + 1\right), N\right) \text{ or } \text{GCD}\left(\left(a^{\frac{r}{2}} - 1\right), N\right)$$

Using Period to Get Factors

- Problem Can Occur if: $a^{\frac{r}{2}} \equiv -1 \pmod{N}$
- When This Occurs Right Side of Following Equation Becomes Zero and no Information about N Results

$$N \mid \left(a^{\frac{r}{2}} + 1 \right) \left(a^{\frac{r}{2}} - 1 \right)$$

- If This Occurs Must Try Again With Different Value of a

GCD and Factor Example

- Period of $f_{2,15}$ is 4 or: $2^4 \equiv 1 \pmod{15}$

x	0	1	2	3	4	5	6	7	8	9	10	11	12	...
$f_{2,15}(x)$	1	2	4	8	1	2	4	8	1	2	4	8	1	...

- Using Previous Result with GCD:

$$15 \mid (2^2 + 1)(2^2 - 1)$$

$$\text{GCD}(5, 15) = 5$$

$$\text{GCD}(3, 15) = 3$$

GCD and Factor Example

- Period of $f_{6,371}$ is 26 or: $6^{26} \equiv |1|_{371}$

x	0	1	2	3	4	5	6	7	...	13	...	25	26	27	28	...
$f_{6,371}(x)$	1	6	36	216	183	356	281	202	...	370	...	62	1	6	36	...

- It Is Also True That:

$$6^{\frac{26}{2}} = 6^{13} \equiv |370|_{371} \equiv |-1|_{371}$$

- This is the Problem Case
- Must Discard $a=6$ and Try Again With New a Value

GCD and Factor Example

- Period of $f_{24,371}$ is 78 or $24^{78} \equiv 1 \pmod{371}$

x	0	1	2	3	4	5	6	7	...	39	...	77	78	79	80	...
$f_{24,371}(x)$	1	24	205	97	102	222	134	248	...	160	...	201	1	24	205	...

- Checking for Problem Case:

$$24^{\frac{78}{2}} = 24^{39} \equiv 160 \pmod{371} \neq -1 \pmod{371}$$

- Can Use This a Value

$$371 \mid (24^{39} + 1)(24^{39} - 1)$$

$$\text{GCD}(161, 371) = 7 \qquad \text{GCD}(159, 371) = 53$$

$$371 = 7 \times 53$$

Shor's Factoring Algorithm

Input: Positive Integer N with $n = \lceil \log_2(N) \rceil$

Output: Factor p of N if it Exists

- 1) Use classical polynomial algorithm to determine if N is prime or a power of a prime. If N is prime or power of prime, declare that it is and halt.
- 2) Randomly choose an integer a such that $1 < a < N$. Invoke Euclid's algorithm to determine $\text{GCD}(a, N)$. If GCD is not 1, Then Halt.

Shor's Factoring Algorithm

- 3) Use Quantum Circuit to find period r .
- 4) If r is odd or is the “problem case”, return to step 2 and choose another a value.
- 5) Invoke Euclid's algorithm to calculate:
$$\text{GCD}\left(\left(a^{\frac{r}{2}} + 1\right), N\right) \text{ or } \text{GCD}\left(\left(a^{\frac{r}{2}} - 1\right), N\right)$$

Return at least one of the nontrivial solutions.

Implementing $Uf_{a,N}$

- Operation of $f_{a,N}(\mathbf{x})$ Considered on Bit-by-Bit Basis
- Radix Polynomial Representation of $\mathbf{x}$:

$$\mathbf{x} = x_{n-1}2^{n-1} + x_{n-2}2^{n-2} + \dots + x_22^2 + x_12^1 + x_02^0$$

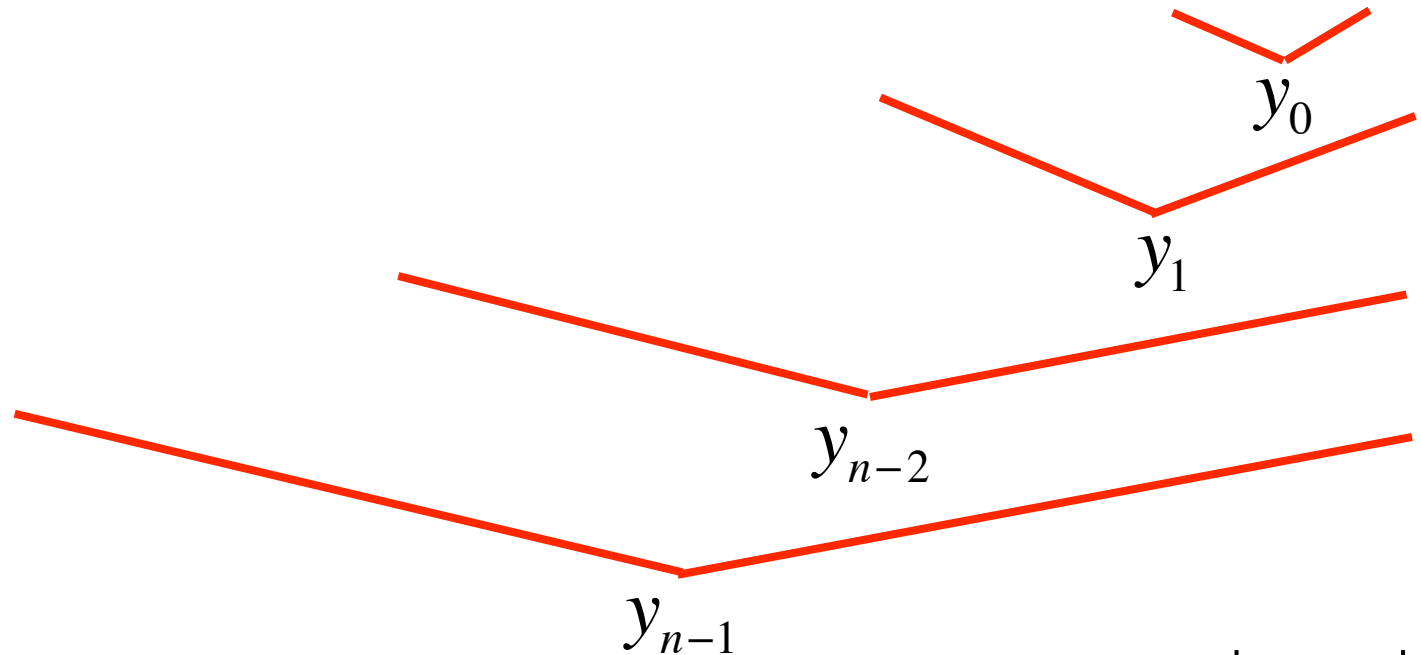
$$f_{a,N}(\mathbf{x}) = \left| a^{\mathbf{x}} \right|_N = \left| a^{x_{n-1}2^{n-1} + x_{n-2}2^{n-2} + \dots + x_22^2 + x_12^1 + x_02^0} \right|_N$$

$$f_{a,N}(\mathbf{x}) = \left| a^{x_{n-1}2^{n-1}} \right|_N \times \left| a^{x_{n-2}2^{n-2}} \right|_N \times \dots \times \left| a^{x_12^1} \right|_N \times \left| a^{x_02^0} \right|_N$$

Implementing $Uf_{a,N}$

- Rewrite This as an Inductive Formula:

$$f_{a,N}(\mathbf{x}) = \left| a^{x_{n-1} 2^{n-1}} \right|_N \times \left| a^{x_{n-2} 2^{n-2}} \right|_N \times \dots \times \left| a^{x_1 2^1} \right|_N \times \left| a^{x_0 2^0} \right|_N$$



$$f_{a,N}(\mathbf{x}) = y_{n-1} = y_{n-2} \times \left| a^{x_{n-1} 2^{n-1}} \right|_N \quad y_j = y_{j-1} \times \left| a^{x_j 2^j} \right|_N$$

Implementing $Uf_{a,N}$

$$y_j = y_{j-1} \times \left| a^{x_j 2^j} \right|_N$$

- When $x_j=0$ We Have:

$$y_j = y_{j-1}$$

- When $x_j=1$ We Should Multiply y_{j-1} by:

$$\left| a^{x_j 2^j} \right|_N$$

- When a and N are Coprime, Operation of Multiplying by This Factor is Reversible and Unitary – Realizable as Quantum Cascade

Implementing $U_{f_{a,N}}$

- For Each j , There is a Unitary Operator:

$$U_{\left|a^{2^j}\right|_N} \rightarrow U_{a^{2^j}}$$

- Each of These Operators are Performed Conditionally Based on Value of x_j
- To Implement we use a Controlled Version of the Operator
- The Quantum Cascade has the Form as Shown on the Following Overhead

Implementing $U_{f_{a,N}}$

